

(SELV) power in the vehicle. It provides internal isolation between high and low voltage, which reduces the space needed for DC-DC conversion. With a power density of 158kW/L, it helps EV system designers shrink the primary DC-DC converter and lower vehicle weight. Its bidirectional current response rate of 8 mega amps per second enables it to act as a virtual 48V battery, replacing a 25lb physical battery and reducing both cost and weight.

Vicor Power

<https://www.vicorpower.com>

LoRa transceiver for AI-enabled edge devices

Semtech Corporation has introduced the LR2021, the first chip in the LoRa Plus family. The transceiver, which incorporates fourth-generation LoRa



IP, supports both terrestrial and SATCOM networks in the sub-GHz, 2.4GHz ISM bands, and licensed S-band. It is back-

wards compatible with previous LoRa devices, ensuring LoRaWAN compatibility. Designed for AI-enabled edge devices, the LR2021 supports sensor data transmission for applications like gunshot detection, glass break sensing, license plate recognition, and fall detection, while running on small batteries for years. With a data rate of up to 2.6Mbps, it supports audio streaming and image transfer. The RF and analogue architecture supports multi-region deployment through a single-SKU design, while its integration reduces BOM costs, PCB footprint, and power consumption compared to previous LoRa transceivers.

Semtech Corporation

<https://www.semtech.com>

A miniature precision lidar depth sensor

Sony has introduced the AS-DT1, a miniature precision lidar depth sensor. This sensor, measuring 29 by 29 by 31 millimetres, excluding

protrusions, uses 'Direct Time of Flight' (dToF) lidar technology and features a Sony Single Photon



Avalanche Diode (SPAD) image sensor. The sensor's small size and SPAD sensor make it suitable for applica-

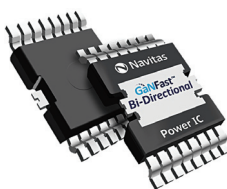
tions where space and weight are critical, such as drones, robotics, and self-driving vehicles. Its design allows for integration into devices like food service robots, autonomous mobile robots, and drones used for inspections and surveys. The sensor functions in both indoor and outdoor environments, providing accurate measurements in different lighting conditions.

Sony

<https://www.sony.co.in/homepage>

650V bi-directional GaNFast and IsoFast gate driver ICs

Navitas Semiconductor has launched the first 650V bi-directional GaNFast ICs and IsoFast gate driver ICs. These



ICs change how power systems are designed by allowing a shift from two-stage to single-stage setups. They

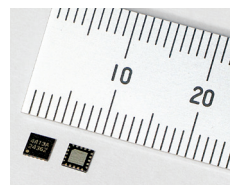
are used in areas such as EV charging (on-board and roadside), solar inverters, energy storage systems, and motor drives. Bi-directional GaNFast power ICs function like two GaN switches placed back-to-back. They are single-chip designs with a shared drain, two gate controls, and a built-in substrate clamp. This technology enables single-stage bi-directional switch (BDS) converters, supporting the move from two-stage to single-stage designs. This leads to improved efficiency, power density, performance, and reduced system cost and complexity.

Navitas Semiconductor

<https://navitassemi.com>

Low-current PMICs for battery charging

Asahi Kasei Microdevices (AKM) has released the AP4413, a series of low-current power management ICs (PMICs) for battery charging in energy harvest-



ing systems. It consumes 52nA of current and charges batteries using low amounts of energy. It comes

in four versions with voltage thresholds for different rechargeable battery types. The AP4413 includes AKM's low-current voltage monitoring system, allowing it to charge batteries using harvested energy. This supports devices like remote controls, Bluetooth trackers, and IoT sensors, which use disposable batteries. The PMIC reduces its energy use and protects batteries from over-charging or deep discharge.

Asahi Kasei Microdevices (AKM)

<https://www.akm.com/global/en>

Battery made from aloe vera for a greener future

Aloe E-Cell has developed a battery made from aloe vera. This reduces e-waste, supports farmers, and involves women in clean technology. It also follows a cir-



cular economy model by turning used batteries into plant fertilisers. It replaces 60% of the

toxic materials used in regular batteries. Aloe vera works in batteries as an electrolyte and binder. Aloe E-Cell uses no chemicals like mercury or cadmium, replaces plastic with materials like paper, and lasts up to 1.5 times longer than regular batteries. Made from plant-based ingredients, it is safe for the environment and renewable. These batteries also have self-regenerating properties, reducing the need for additives.

Aloe E-Cell

<https://www.aloeecell.com>